

MDS-UPDRS Finger Tapping Score

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested.

Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [150, 149, 154, 172, 156, 183, 138, 170, 197, 131]

Tap amplitudes (pixels): [122.41323457861898, 118.81077392223317, 110.43550153822818, 114.85643212288984, 107.07474025184464, 107.07474025184464, 104.04326023342406, 116.21101496846157, 109.85900054160332, 136.12494260788506, 98.71676655968832]

GPT-5

MDS-UPDRS Item 3.4 Finger Tapping Scoring Analysis

What's given

- Instruction target: 10 consecutive taps
- Extracted:
 - Number of taps detected: 11
 - Tap periods (frames) for 10 intervals: [150, 149, 154, 172, 156, 183, 138, 170, 197, 131]
 - Tap amplitudes (pixels) for 11 taps: [122.41, 118.81, 110.44, 114.86, 107.07, 107.07, 104.04, 116.21, 109.86, 136.12, 98.72]

Note: 11 amplitude values corresponds to 11 detected taps; 10 periods correspond to the intervals between the first 11 taps. Scoring is based on the first 10 taps per the test instruction; the extra tap likely reflects an over-tap at the end.

Speed (periods) assessment

- Mean period $\approx (150+149+154+172+156+183+138+170+197+131)/10 = 160.0$ frames (moderately variable).
- Variability: noticeable spread from 131 to 197 frames; several longer intervals (172, 183, 170, 197) interspersed with shorter ones (131–156).

- Interpretation:
 - Not uniformly slow, but there are multiple hesitations/irregularities indicated by longer periods and variability.
 - This pattern is more than “slight slowing.” It suggests repeated interruptions/hesitations rather than a single brief break.

Rhythm/interruptions

- Using period irregularities as a proxy for hesitations/interruptions:
 - At least 5 intervals stand out as prolonged vs adjacent ones (172, 156–183 cluster, 170, 197), indicating multiple hesitations.
 - This suggests ≥ 3 interruptions; plausibly >5 smaller rhythm breaks.
- This aligns with:
 - “3 to 5 interruptions” → score 2 (Mild), or
 - “More than 5 interruptions or at least one longer arrest (freeze)” → score 3 (Moderate).
- No explicit freeze/arrest is evident (no single period outlier suggesting a true halt), but the number of rhythm breaks appears to exceed 5.

Amplitude assessment

- Amplitudes by tap (1 → 11): 122.4, 118.8, 110.4, 114.9, 107.1, 107.1, 104.0, 116.2, 109.9, 136.1, 98.7
- Pattern:
 - Early decline from tap 1 → 3, slight recovery at 4, further decline to 7, recovery at 8, moderate at 9, large spike at 10, then drop at 11.
 - Decrement is not monotonic; it begins early (after the 1st tap) but is inconsistent with rebounds.
- Mapping to criteria:
 - “Decrements near the end” → score 1
 - “Decrements midway” → score 2
 - “Decrements starting after the 1st tap” → score 3
- Given the early decline from tap 1 to 3 and continued lower amplitudes through mid-sequence (even with rebounds), this matches “decrements starting after the 1st tap,” supporting score 3.

Overall synthesis

- Speed/rhythm: multiple hesitations likely >5 minor interruptions → leans toward score 3.
- Amplitude: decrement begins after the first tap → score 3.
- The task is clearly performed (not barely or unable), so score 4 (severe) is not supported.

Final score: 3
